

**Program:**

```
import nltk

from nltk.tag import hmm

from nltk.corpus import treebank


# Download required NLTK resources

nltk.download('treebank')

nltk.download('universal_tagset')

# Load a portion of the Penn Treebank corpus

train_data = treebank.tagged_sents(tagset='universal')[:3000]

test_data = treebank.tagged_sents(tagset='universal')[3000:]

# Train an HMM model

trainer = hmm.HiddenMarkovModelTrainer()

hmm_tagger = trainer.train(train_data)

# Test the model

accuracy = hmm_tagger.evaluate(test_data)

print(f'HMM Tagger Accuracy: {accuracy:.4f}')


# Test with a sentence

sentence = "The quick brown fox jumps over the lazy dog".split()

print("Tagged Sentence:")

print(hmm_tagger.tag(sentence))
```

**Output:**

Tagged Sentence:

[('The', 'DET'), ('quick', 'ADJ'), ('brown', 'ADJ'), ('fox', 'NOUN'), ('jumps', 'VERB'), ('over', 'ADP'), ('the', 'DET'), ('lazy', 'ADJ'), ('dog', 'NOUN')]